## 11.5 部分习题参考简答

1. (1) $\frac{dy}{dx}=\frac{e^y}{1-xe^y}$；　(2) $\frac{dy}{dx}=\frac{\cos x}{2\sin y}$.

2. (1) $\frac{\partial z}{\partial x}=\frac{1+yz\sin(xyz)}{1-xy\sin(xyz)}$, $\frac{\partial z}{\partial y}=\frac{1+xz\sin(xyz)}{1-xy\sin(xyz)}$.

(2) $\frac{\partial z}{\partial x}=\frac{x^2-yz}{xy-z^2}$, $\frac{\partial z}{\partial y}=\frac{y^2-xz}{xy-z^2}$.

3.（1）在 $\mathbb{R}^2$ 上的任一点附近都可确定 $z=z\left(x,y\right)$，且 $\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}=-1$; $\frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}=-1$.

（2）除了$\begin{cases} y\cos yz+x\cos xz=0 \\ \sin xy+\sin yz+\sin zx=0 \end{cases}$上的点，其余点附近可确定 $z=(x,y)$,

且 $\frac{\partial z}{\partial x}=-\frac{y\cos xy+z\cos xz}{y\cos yz+x\cos xz}$, $\frac{\partial z}{\partial y}=-\frac{x\cos xy+z\cos yz}{y\cos yz+x\cos xz}$.

4.　（1） $c\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}=a$.

（2） $x\frac{\partial z}{\partial x}+z\frac{\partial z}{\partial y}=y$.

（3） $\frac{\partial z}{\partial x}=-(1+\frac{f_1'+f_2'}{f_3'})$， $\frac{\partial z}{\partial y}=-(1+\frac{f_2'}{f_3'})$，

$$\frac{\partial^2 z}{\partial x^2}=-\frac{f_{11}''+f_{12}''+f_{21}''+f_{22}''}{(f_3')^2}+\frac{(f_1'+f_2')(f_{13}''+f_{23}'')}{(f_3')^3}.$$

（4） $\frac{\partial z}{\partial x}=\frac{-y}{x^2+y^2}$， $\frac{\partial z}{\partial x}=\frac{x}{x^2+y^2}$， $\frac{\partial^2 z}{\partial x^2}=\frac{2xy}{(x^2+y^2)^2}$.

5. 在 $P(-1,1,0)$ 附近能确定 $\begin{pmatrix} y \\ z \end{pmatrix}=\mathbf{f}\left(x\right)$，且 $y'\left(-1\right)=0,\quad z'\left(-1\right)=-1$.

6. $dz=-\frac{f_1'+2xf_2'}{f_1'+2zf_2'}dx-\frac{f_1'+2yf_2'}{f_1'+2zf_2'}dy.$

7. $\frac{\partial^2 z}{\partial x\partial y}=\frac{9x^2y^2\left(6z-\varphi''(z)\right)}{\left(\varphi'(z)-3z^2\right)^3}$.

10. $\dfrac{\partial^2 w}{\partial u^2}+\dfrac{\partial w}{\partial u}=0.$

11. 令 $\mathbf{x}_0=(0,\dfrac{1}{2},\sqrt{\dfrac{1}{8}},\sqrt{\dfrac{5}{8}})$，则 $\left.\dfrac{\partial(u,v)}{\partial(x,y)}\right|_{(0,\frac{1}{2})}=\begin{pmatrix} 0 & \dfrac{\sqrt{2}}{2} \\ 0 & -\dfrac{3\sqrt{10}}{10} \end{pmatrix}$，

其微分为 $\mathrm{d}f\left(0,\dfrac{1}{2}\right)=\begin{pmatrix} \dfrac{\sqrt{2}}{2}\mathrm{d}y \\ -\dfrac{3\sqrt{10}}{10}\mathrm{d}y \end{pmatrix}.$

12．(1) $\dfrac{\partial(x,y)}{\partial(u,v)}=\dfrac{1}{4\left(x^2+y^2\right)}\begin{pmatrix} 2x & 2y \\ -2y & 2x \end{pmatrix}$，其行列式为 $\dfrac{1}{4\left(x^2+y^2\right)}$；

(2) $\dfrac{\partial(x,y)}{\partial(u,v)}=\begin{pmatrix} e^{-x}\cos y & e^{-x}\sin y \\ -e^{-x}\sin y & e^{-x}\cos y \end{pmatrix}$，其行列式为 $e^{-2x}$；

(3) $\dfrac{\partial(x,y)}{\partial(u,v)}=\dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$，其行列式为 $\dfrac{1}{ad-bc}$；

(4) $\dfrac{\partial(x,y)}{\partial(u,v)}=\dfrac{1}{6x^3y+3y^4}\begin{pmatrix} 2xy & 3y^2 \\ -y^2 & 3x^2 \end{pmatrix}$，其行列式等于 $\dfrac{1}{6x^3y+3y^4}$.

13. $\dfrac{\partial\left(\rho,\varphi\right)}{\partial\left(x,y\right)}=\dfrac{1}{\rho}\begin{pmatrix} \rho\cos\varphi & \rho\sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix}$.

14. $\dfrac{\partial u}{\partial x}(3,4)=2$.